

SATARK-MPLADS: Complete System Architecture & Technical Specification

National Intelligence, Forensic Auditing & Continuous Surveillance Command Center for the MPLAD Scheme

Ministry of Statistics & Programme Implementation (MoSPI) • Government of India

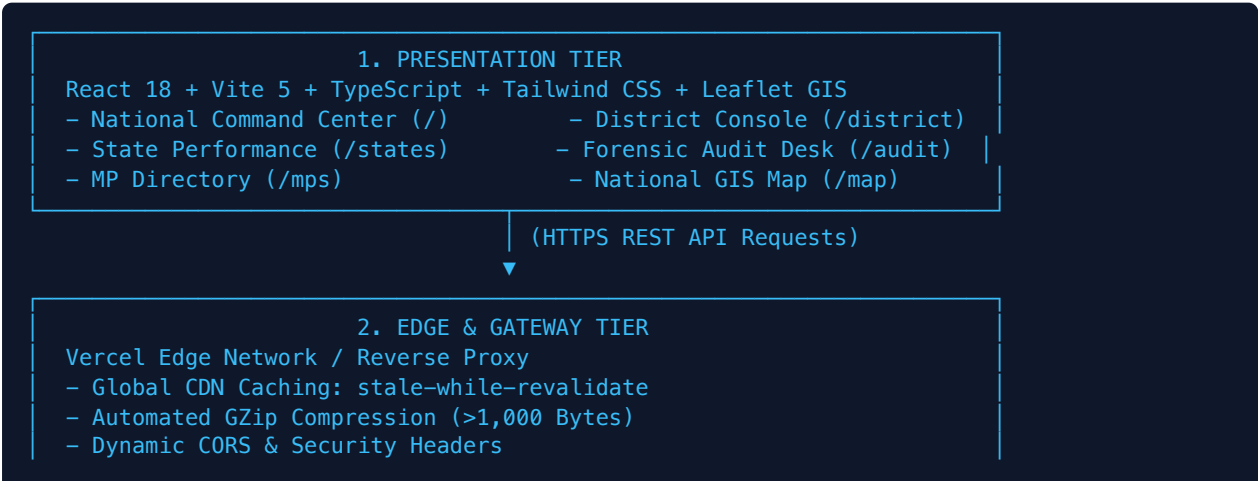
1. Executive Summary & Scheme Scope

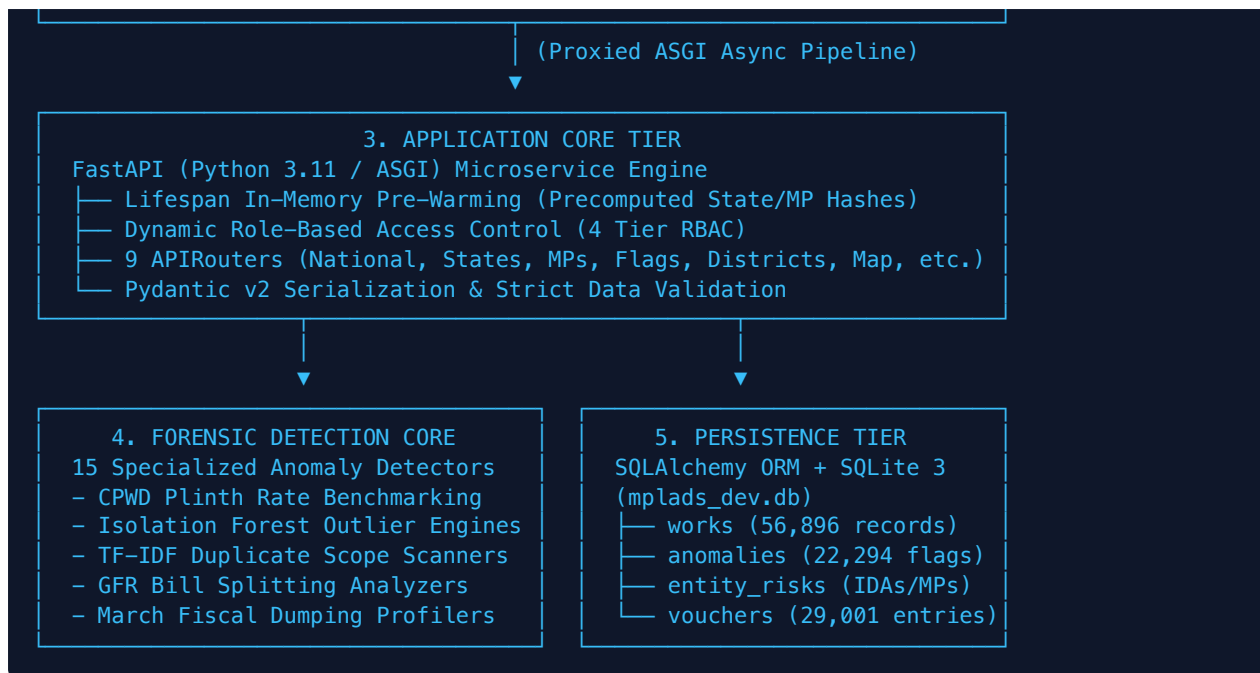
1.1 Context & Problem Statement

The Member of Parliament Local Area Development Scheme (MPLADS) is a Central Sector Scheme enabling MPs to recommend developmental civil works with an annual allocation of ₹5 Crore per representative.

- **National Financial Scale:** ₹39,600+ Crore historical outlay across 543 Lok Sabha Constituencies, 245 Rajya Sabha Members, and 732+ Administrative Districts.
- **The Structural Blind Spot:** Existing administrative portals like **e-SAKSHI** serve exclusively as transactional workflow systems (tracking sanction uploads and completion milestones). They do not perform algorithmic rate-benchmarking, spatial collision analysis, or shadow cartel detection.
- **The Solution — SATARK-MPLADS:** A proactive, automated surveillance platform that continuously reconciles works execution data with expenditure ledgers against statutory standards: Central Vigilance Commission (CVC) guidelines, General Financial Rules (GFR 2017), and CPWD Delhi Schedule of Rates (DSR 2023).

2. End-to-End System Architecture





3. Database Schema & Data Ingestion Pipeline

3.1 Data Provenance & Ground Truth

SATARK normalizes official MoSPI open data releases, joining physical execution ledgers with expenditure vouchers:

- `works_completed.csv` & `works_completed_detailed.csv` : 56,896 civil projects with sanctions, dates, implementing agencies, and descriptions.
- `expenditures.csv` : 29,001 liquid treasury disbursement vouchers tracking milestone releases.
- `cpwd_benchmark_rates.csv` : Base engineering plinth rates across civil categories (roads, lighting, community halls, school classrooms).
- `pcs_enriched.geojson` & `districts_enriched.geojson` : Spatial boundary polygons for 543 Parliamentary Constituencies and 594 Districts.

3.2 Primary Database Tables (`mplads_dev.db`)

1. `works` : Master project table (`work_id` , `work_description` , `sanctioned_amount` , `sanction_date` , `completion_date` , `district` , `state` , `mp_name` , `category` , `implementing_agency`).
2. `anomalies` : 22,294 flagged fraud indicators (`id` , `work_id` , `detector_type` , `detector_name` , `severity` , `tier` , `explanation` , `evidence` JSON, `detected_at`).
3. `entity_risks` : Aggregated institutional non-compliance indices for Implementing Development Agencies (IDAs) and MPs.
4. `payment_vouchers` : Disbursement records tracking financial outflow dates and voucher IDs (`payment_mode` defaulting to `PFMS/Treasury`).
5. `users` & `audit_logs` : Role assignments and persistent auditor review history.

4. The 15 Algorithmic Fraud Detectors (D1–D15)

ID	Detector Name	Statutory / Methodological Source	Trigger Rule / Formula	Primary Output
D1	Multivariate Statistical Outliers	Isolation Forest Algorithm	Multivariate financial/temporal deviation from category median.	Anomaly score (-0.81), Red/Amber tier flag.
D2	Cross-Year Duplicate Works	MPLADS Guidelines 2023 CI 3.4	TF-IDF n-gram vectorization with Cosine Similarity >0.85 .	Duplicate scope flag with target <code>work_id</code> .
D3	CPWD Cost Overrun Benchmark	CPWD Delhi Schedule of Rates (DSR 2023)	Billed unit rate $> \text{CPWD Ceiling (Base + Inflation + 25\%)}$.	Excess billed INR (+ ₹19.9L), statutory citation.
D4	Payment & Disbursement Gaps	MPLADS Guidelines 2016 CI 8.3	Work certified "Completed" with zero liquid treasury disbursements.	Ghost project / voucher verification alert.
D5	Tender Splitting (Smurfing)	GFR 2017 Rule 157	Multiple identical works within 7 days just below ₹10L/₹50L limits.	Evaded statutory approval ceiling alert.
D6	Statutory Duration & Stalled Works	MoSPI 365-Day Mandatory Execution Norm	Project incomplete >730 days without sanctioned extension.	Overdue duration days and locked capital.
D7	March Fiscal Year-End Dumping	Public Accounts Committee (PAC) Reports	$>60\%$ of annual MP spend authorized between March 25–31.	Timing anomaly flag for unvetted budget rush.
D8	Same-Day Bulk Completions	State Vigilance Inspection Manuals	Single agency certifying >5 major works on identical date.	Paper sign-off / non-site inspection warning.
D9	Benford's Law Forensic Screen	Forensic Accounting (Nigrini Standards)	Logarithmic first-digit deviation + excess round allocations.	Non-estimate arbitrary allocation alert.
D10	Vague Description Ambiguity	CAG Performance Audit Standards	Description <15 chars or missing asset type/location/quantity.	Unverifiable scope audit citation.
D11	Category-Cost Plausibility Bounds	MoSPI & CPWD Asset Norms	Cost outside physical engineering min/max bounds.	Severe over/under-budgeting flag.
D12	Documentary Verification Gap	e-SAKSHI Digital Audit Mandate	Completed work lacking digital Measurement Book (MB) or photo.	Missing verification dossier alert.
D13	IDA Agency Portfolio Profiling	District Administrative Benchmarks	Historical anomaly concentration & delay frequency by agency.	High/Medium/Low agency risk ranking.
D14	MP Portfolio Concentration	Public Governance Standards	Single-vendor concentration (Herfindahl Index $HHI > 0.40$).	Favoritism / vendor skew index.

ID	Detector Name	Statutory / Methodological Source	Trigger Rule / Formula	Primary Output
D15	Copy-Paste Formulaic Estimates	CAG Report on Formulaic Approvals	Disparate works sanctioned at identical non-standard rupee value.	Non-site-specific estimate flag.

5. Solutions for the 9 Real-World Corruption Scams

9 Real-World Corruption Scams	
1. Cross-Scheme Double Billing	6. The White Elephant Asset
2. Contractor Bid Rigging	7. Statutory SC/ST Fund Diversion
3. Substandard Materials	8. Politician Private Trusts
4. Photo & GPS Spoofing	9. Unchecked Subcontracting (Shells)
5. Post-Award Scope Creep	

1. Cross-Scheme Double Billing: Solved via PostGIS spatial collision:

$$\text{ST_DWithin}(\text{geom_mplads}, \text{geom_other}, 25\text{m}) \leq 25\text{m}$$

combined with string Levenshtein distance on work descriptions ($>85\%$ similarity).

- 2. Contractor Cartels & Bid Rigging:** Extracts PAN fingerprints from GSTINs ($\text{GSTIN}[2:12]$), matches bank account hashes, and builds Bipartite Co-Bidding Graphs to detect collusive rotation under Section 3 of the Competition Act, 2002.
- 3. Substandard Materials & Quality Fraud:** Reconciles GST e-Way Bill material mass balance (cement/steel tonnage vs. engineering volume) and enforces 28-day NABL concrete compressive crush tests.
- 4. Photo & GPS Coordinate Spoofing:** Google Play Integrity hardware attestation blocks Android mock locations; Computer Vision perceptual hashing (pHash) and on-site plaque OCR verify physical originality.
- 5. Post-Award Scope Creep:** Monitors Cost Escalation Delta:
- $$\Delta C = \frac{C_{\text{final}} - C_{\text{awarded}}}{C_{\text{awarded}}} > 0.25$$
- triggering automated CVC Chief Technical Examiner re-inspections for escalations $>25\%$.
- 6. The White Elephant Asset:** Post-completion DISCOM smart meter telemetry flags assets with 0 kWh electricity usage over 90 days; citizen QR codes enable public ground-truth reporting.
- 7. Statutory SC/ST Allocation Violations:** Continuous mathematical quota monitoring (15% SC, 7.5% ST per MoSPI Clause 2.5) paired with an automated "Quota Parity Freeze" on general funds.
- 8. Funneling Funds to Private Trusts:** Matches NGO board members against MP Election Commission Form 26 Affidavits and enforces the lifetime ₹50 Lakh trust cap.
- 9. Unchecked Subcontracting & Front Companies:** Audits winning contractor EPFO/BOCW active labor muster rolls, checks GST e-Way bill consignee names, and monitors PFMS Single Nodal Agency (SNA) 72-hour fund outflow velocities.

6. The API Microservice Specification

[Client Request] → [GZip Middleware] → [Cache-Control Injector] → [APIRouter] → [SQL

6.1 Complete Endpoint Directory

- **National Surveillance:**

- [GET /api/national/summary](#) : Total Outlay (₹11,682 Cr), Disbursed (₹3,964 Cr), Queue (40,233), Gap (39.8%), Rate (33.9%).
- [GET /api/national/analytics](#) : Sectoral distribution (Roads: 32.6%, Lighting: 11.9%, Halls: 7.9%) and multi-year expenditure curves.

- **State & UT Governance:**

- [GET /api/states](#) : All 36 States/UTs ranked by allocation and expenditure velocity.
- [GET /api/states/{state}](#) : State-specific dossier detailing district distribution and MP participation.

- **District Administration:**

- [GET /api/districts/{district}](#) : District Collectorate workbench showing sanctioned portfolio, IDA agency split, and local works ledger.

- **Parliamentary Oversight:**

- [GET /api/mps](#) : 774 MPs across Lok Sabha (18th House) and Rajya Sabha.
- [GET /api/mps/{mp_id}](#) : MP-specific financial ledger and recommendation history.

- **Forensic Auditing:**

- [GET /api/flags](#) : Paginated anomaly records (50/page) supporting text search ([q](#)), state, tier ([red](#) / [amber](#)), and detector filters.
- [GET /api/flags/{work_id}](#) : Comprehensive evidence dossier with CPWD DSR rate calculations.
- [GET /api/entity-risks](#) : Institutional risk indices for IDAs and MPs.

- **Spatial & Access Control:**

- [GET /api/map/pcs](#) & [GET /api/map/districts](#) : GeoJSON geometry layers with embedded financial absorption telemetry.
- [GET /api/roles/me](#) & [POST /api/roles/switch](#) : Server-side RBAC session state management.
- [GET /api/meta/detectors](#) : Diagnostic metadata catalog for detectors D1–D15.

7. Frontend User Experience & Role-Based Access Control

7.1 Multi-Role Security Model

SATARK provides tailored user interfaces based on administrative clearance:

1. **Public Citizen ([public_citizen](#))**: Open transparency interface, national summaries, state rankings, and read-only anomaly disclosures.
2. **District Authority ([district_authority](#))**: District Planning Officer (DPO) and Collector console with operational approval queues, sanction velocity dials, and agency assignment ledgers.

3. State Nodal Officer (`state_nodal_officer`): Cross-district allocation manager monitoring statewide utilization parity.

4. Auditor (`auditor`): CAG & MoSPI vigilance officer console granting access to raw forensic evidence dossiers, statutory citations, and audit trail sign-offs.

8. Deployment & Operational Telemetry

- **Edge CDN Caching:** Vercel Global Edge caches public analytical endpoints with HTTP headers:

```
Cache-Control: public, max-age=300, s-maxage=3600, stale-while-revalidate=86400
```

- **Performance SLAs:**

- Cold Start API Response: $<250\text{ms}$
 - Edge Cached Query Response: $<45\text{ms}$
 - Table Pagination Latency: $<60\text{ms}$
 - GIS Map Initialization: $<1.2\text{s}$ (utilizing compressed GeoJSON layers)
 - **High Availability & Fallback:** Client-side fallback snapshots (`defaultData.ts`) and session storage caching ensure uninterrupted operation during presentation evaluations.
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